

# AIVA

## AI-Powered Interview Evaluation Assistant

*Artificial Intelligence Interview Validation Assistant*

### 01 Project Definition

AIVA is an AI-assisted recruitment evaluation platform designed to support interviewers throughout the hiring process. Unlike traditional ATS systems that rely on keyword matching, AIVA evaluates candidate evidence, technical performance, problem-solving ability, coding efficiency, and interview responses — while keeping the final hiring decision under human control.

The system acts as an **intelligent evaluator assistant** rather than an autonomous recruiter. It provides objective metrics, verification reports, statistical insights, and candidate assessments to help interviewers make more reliable and data-driven hiring decisions.

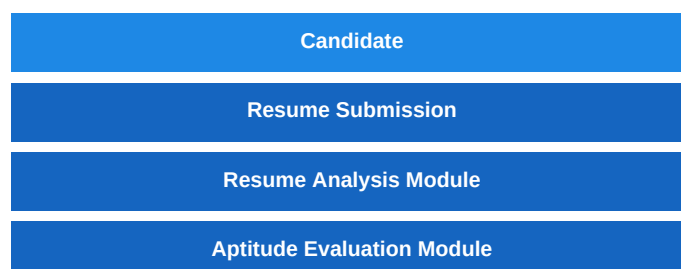
### 02 Problem Statement

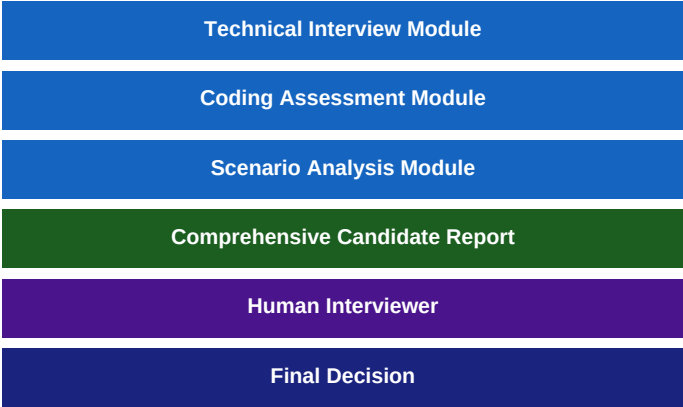
- Resumes can contain exaggerated or unverifiable claims.
- Recruiters often spend less than a minute reviewing resumes.
- Candidate evaluation varies significantly between interviewers.
- Coding assessments focus mainly on correctness rather than efficiency.
- Technical interviews lack objective measurement standards.
- Human bias can affect candidate rankings.

### 03 Project Objectives

- ✓ Verify resume claims with evidence-based analysis.
- ✓ Evaluate aptitude performance through structured testing.
- ✓ Assess technical interview responses in real time.
- ✓ Analyze coding efficiency (time & space complexity).
- ✓ Generate comprehensive interviewer support reports.
- ✓ Provide statistical evidence for candidate comparison.
- ✓ Keep final hiring decisions under human control.

### 04 Overall Workflow





## 05 System Modules

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module 1 | <b>Resume Intelligence Engine</b> <p><b>Purpose:</b> Evaluate candidate qualifications and verify claims against real evidence.</p> <ul style="list-style-type: none"><li>• Resume Parsing – Extracts name, education, skills, projects, experience, certifications.</li><li>• Skill Matching – Compares job requirements vs candidate skills; generates match percentage.</li><li>• Hyperlink Verification – Validates GitHub repos, LinkedIn profiles, portfolio websites, certificates.</li><li>• Evidence-Based Verification – Checks actual GitHub Python projects instead of just the keyword 'Python'.</li></ul> <p><b>Sample Output:</b></p> <div>Resume Score: 85 / 100</div> <div>Skill Match: 90%   Evidence Strength: High</div> <div>Project Verification: Verified   Certification: Verified</div> |
| Module 2 | <b>Aptitude Evaluation Engine</b> <p><b>Purpose:</b> Measure efficiency, accuracy, and reasoning ability across three domains.</p> <ul style="list-style-type: none"><li>• Quantitative Aptitude – Mathematics, numerical reasoning.</li><li>• Logical Reasoning – Pattern recognition, analytical thinking.</li><li>• Verbal Ability – Language understanding, communication basics.</li><li>• Metrics: Accuracy rate, average response time, difficulty index (Easy / Medium / Hard).</li></ul> <p><b>Sample Output:</b></p> <div>Accuracy: 92%</div> <div>Avg Time – Easy: 12 sec   Medium: 35 sec   Hard: 70 sec</div> <div>Problem Solving Index: High</div>                                                                                                                                                |
| Module 3 | <b>Technical Interview Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Purpose:** Support the interviewer during technical discussions with AI-generated questions.

- AI Question Generation – Generates questions from candidate projects, skills, and degree.
- Example: Project 'Navis' → Why weighted routing? How is route confidence calculated?
- Real-Time Analysis – Monitors technical accuracy, concept understanding, and consistency.

**Sample Output:**

Technical Depth: 8.5 / 10

Project Understanding: High | Consistency Score: 9 / 10

Module 4

Coding Assessment Engine

**Purpose:** Evaluate coding skills objectively using correctness, complexity, and quality metrics.

- Correctness – Runs public + hidden test cases.
- Time Complexity Analysis – Estimates  $O(1)$ ,  $O(\log n)$ ,  $O(n)$ ,  $O(n \log n)$ ,  $O(n^2)$ .
- Space Complexity Analysis – Estimates memory usage.
- Code Quality – Evaluates readability, structure, and maintainability.
- Benchmark Comparison – Compares against standard and optimal solutions.

**Sample Output:**

Correctness: 100% | Time Complexity:  $O(n \log n)$  | Optimal:  $O(n \log n)$

Space Complexity:  $O(n)$  | Code Quality: 8.7 / 10

Module 5

Scenario-Based Evaluation

**Purpose:** Evaluate real-world engineering thinking through open-ended problem scenarios.

- Example: 'Model accuracy dropped from 95% to 70%. What will you do?'
- AI evaluates: Feasibility, Completeness, Technical accuracy, Risk awareness, Alternatives.

**Sample Output:**

Feasibility: 9/10 | Completeness: 8/10 | Risk Analysis: 8/10

Overall Scenario Score: 8.4 / 10

Module 6

Interviewer Preference Learning

**Purpose:** Learn how experienced interviewers evaluate depth of understanding.

- The AI does NOT learn what is correct or incorrect — it already determines technical correctness independently.
- Instead it learns: what indicates deep understanding vs superficial knowledge.
- Example: 'Random Forest prevents overfitting' → Needs more explanation.
- Example: 'Random Forest reduces variance through bootstrap aggregation' → Strong understanding.
- Outcome: The model learns that depth of explanation matters.

| 06 Comprehensive Candidate Report |       |
|-----------------------------------|-------|
| Evaluation Area                   | Score |

|                     |          |
|---------------------|----------|
| Resume Evaluation   | 88 / 100 |
| Aptitude            | 91 / 100 |
| Technical Knowledge | 85 / 100 |
| Problem Solving     | 90 / 100 |
| Coding Efficiency   | 94 / 100 |
| Scenario Analysis   | 87 / 100 |

|                        |                                          |
|------------------------|------------------------------------------|
| Overall Recommendation | Strong Candidate — Confidence Level: 92% |
|------------------------|------------------------------------------|

## 07 Key Innovation

|                                                                                    |                                                                                                                                                                   |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Traditional ATS asks:</b><br><br>Does the candidate have the required keywords? | <b>AIVA asks:</b><br><br>Can the candidate <b>prove</b> their skills, solve problems efficiently, explain their reasoning, and demonstrate genuine understanding? |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

*AIVA does not replace recruiters or interviewers. It functions as an **AI-powered decision-support platform** that provides objective evidence, statistical analysis, and technical insights to improve hiring reliability and consistency.*